

Annual PM25 pollution surface

This report explains the source data, executed workflow and interpretation of annual pm25 pollution surface. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (ACAG, n.d.; Shen et al., 2024).

Map P11 | Documentation date 6 October 2026 | Units $\mu\text{g}/\text{m}^3$ | Air year 2021

Annual PM2.5 pollution surface

Annual 2021 · $\mu\text{g}/\text{m}^3$

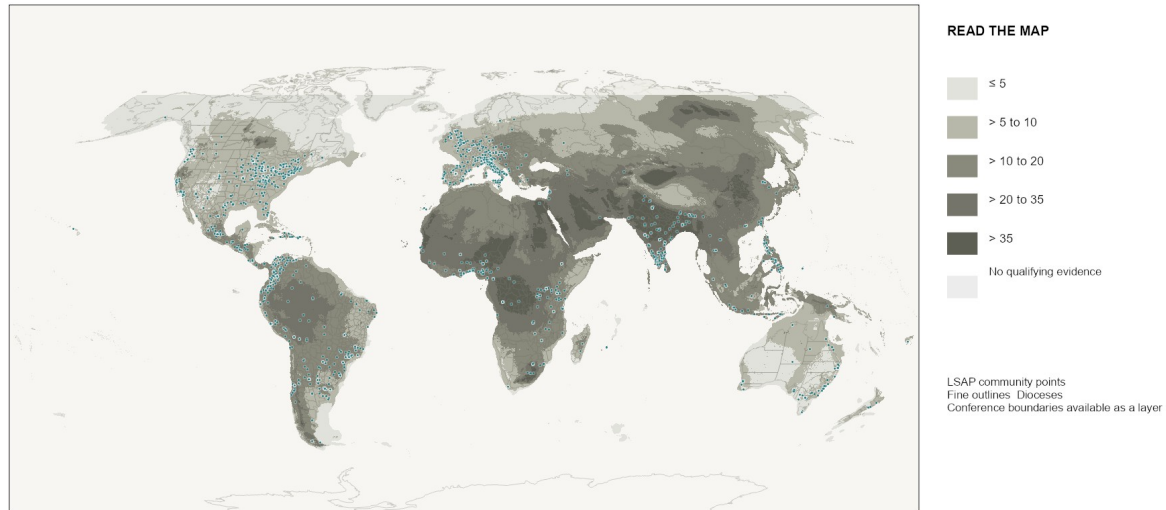


Figure P11 World overview exported from the completed ArcGIS Pro layout.

Native annual concentration surface; individual cells are displayed without the diocesan 50 percent coverage display rule.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (ACAG, n.d.; Shen et al., 2024).

1. Download SatPM V6GL03 CNN global annual 2021 NetCDF; preserve source and attribution.
2. Read native latitude/longitude coordinates and PM25 variable; export a float GeoTIFF on its native grid, units $\mu\text{g}/\text{m}^3$. Convert negative scaled source fill values to NoData in PM25_validated.tif before aggregation and display.
3. Apply the shared weighted raster aggregation to each diocesan copy; retain AirMean, AirMedian, ValidAreaPct, ValidCells and AirStory. Correct latitude-extent coverage against the full polygon.
4. Set AirDisplay to AirMean when valid area coverage is at least 50%; otherwise leave null. Join all ecclesiastical context through the copied polygon attributes.
5. Display PM25_validated.tif with breaks at 5, 10, 20 and 35 $\mu\text{g}/\text{m}^3$ above the country backdrop. Overlay copied LSAP points, diocesan outlines and optional conferences. Companion diocesan statistics are in P01; the 50 percent diocesan threshold does not mask this native surface.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\Sigma(c_i \times a_i) / \Sigma(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO2 implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
Band 1	Annual concentration on the native 0.01 degree grid
NoData	Negative scaled source fill values excluded

Analysis dataset: Rasters/PM25_validated.tif. Styles/P11.lyrx and Styles/P11.mapx preserve layer and map definitions. Map_Exports/P11_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

2021 is the selected common air comparison year. Class boundaries support cartography; they do not classify annual health compliance. Source: SatPM global annual file.

Credit the Atmospheric Composition Analysis Group for the SatPM V6GL03 data, and retain the source and version citation when sharing the map. The source acquisition record identifies CC BY 4.0. Source files and the validated raster copy remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

Atmospheric Composition Analysis Group. (n.d.). SatPM2.5 V6GL03 CNN annual global PM2.5, 2021 [Data set]. Washington University in St. Louis. Native annual file retrieved 6 October 2026. [Source link](#)

Shen, S., Li, C., van Donkelaar, A., Jacobs, N., Wang, C., & Martin, R. V. (2024). Enhancing global estimation of fine particulate matter concentrations by including geophysical a priori information in deep learning. *ACS ES&T Air*, 1(5), 332–345. Background method paper for the V6 product family. [Source link](#)

Esri. (n.d.). Calculate Geometry Attributes. ArcGIS Pro documentation. Reference for geodesic area measurement. Accessed 6 October 2026. [Source link](#)

LSAP Pollution Atlas. (2026). Internal source copies, executed analysis scripts and validation records [Project records]. LSAP_Pollution.aprx; FoundationCopies.gdb; PollutionAnalysis.gdb; foundation_manifest.json; raster_audit.json; freshwater_aggregate_QA.json; waste_QA.json; cleanup_audit.json; project_QA.json. Project root: Atlas Analyses/Pollution. Internal provenance is distinct from published source citations.